

## 교육용 자료 안내

본 자료는 Darun Color Institute의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

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## Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

# Material Safety Data Sheet (MSDS)

## 1. Chemical Product and Company Identification

| Item            | Information                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------|
| Product Name    | Educational Color YELLOW                                                                                 |
| Recommended Use | Cosmetic ingredient                                                                                      |
| INCI Name       | Iron Oxides(CI 77492), Polyglyceryl-2 Triisostearate, Cetyl Ethylhexanoate, Caprylic/Capric Triglyceride |

## 2. Hazards Identification

| Category                 | Details                                                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category                 | Details                                                                                                                                                                                                                                                                                                                |
| GHS Classification       | Health: H315, H319, H335                                                                                                                                                                                                                                                                                               |
| Hazard Statements        | H315: Causes skin irritation<br>H319: Causes serious eye irritation<br>H335: May cause respiratory irritation                                                                                                                                                                                                          |
| Precautionary Statements | P201: Obtain special instructions before use<br>P261: Avoid breathing dust, fumes, gas, mist, spray<br>P273: Do not release into environment<br>P281: Use personal protective equipment as required<br>P312: Take medical service when uncomfortable<br>P403+P233: Store in well-ventilated place. Keep tightly closed |

## 3. Composition/Information on Ingredients

*\*Composition ratios are provided as approximate ranges for educational purposes*

| Ingredient Name               | CAS No.     | Approximate Content (%) |
|-------------------------------|-------------|-------------------------|
| Iron Oxides(CI 77492)         | 1345-27-3   | 48~52%                  |
| Polyglyceryl-2 Triisostearate | 120486-24-0 | 48~52%                  |
| Cetyl Ethylhexanoate          | 59130-69-7  | 5~10%                   |
| Caprylic/Capric Triglyceride  | 73296-86-3  | 5%                      |

## 4. First Aid Measures

| Route of Exposure | First Aid Procedure                                                                      |
|-------------------|------------------------------------------------------------------------------------------|
| Eye Contact       | Rinse with water for at least 15 minutes. Seek medical attention immediately.            |
| Skin Contact      | Rinse with water for at least 15 minutes. Seek medical attention if irritation persists. |
| Inhalation        | Move to fresh air immediately. If breathing is difficult, seek medical attention.        |

|           |                                                             |
|-----------|-------------------------------------------------------------|
| Ingestion | Do not induce vomiting. Seek medical attention immediately. |
|-----------|-------------------------------------------------------------|

## 5. Fire-Fighting Measures

| Item                         | Information                                                                       |
|------------------------------|-----------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Foam, alcohol-resistant foam, carbon dioxide, water spray, dry powder             |
| Specific Hazards             | Irritative and toxic gases may be released during combustion                      |
| Special Protective Equipment | Use self-contained breathing apparatus and chemical-resistant protective clothing |

## 6. Accidental Release Measures

| Action                   | Description                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| Prevention               | Remove all ignition sources; prevent dust formation; ventilate contaminated area                              |
| Handling Spills          | Small spills: Wash with water<br>Large spills: Contain and dispose according to local regulations             |
| Environmental Protection | Avoid contact with skin, eyes, clothing. Do not allow material to enter waterways, drains, or confined spaces |

## 7. Handling and Storage

| Category | Requirements                                                                                                                                               |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling | Use in well-ventilated area<br>Avoid breathing dust and direct skin/eye contact<br>Prevent dust formation                                                  |
| Storage  | Store in well-ventilated place; keep container tightly closed<br>Store away from heat, ignition sources, incompatible materials<br>Keep away from moisture |

## 8. Exposure Controls/Personal Protection

| Control Type                              | Requirement                                            |
|-------------------------------------------|--------------------------------------------------------|
| Control Parameters (Iron Oxides CI 77492) | TWA: 5 mg/m <sup>3</sup>                               |
| Engineering Controls                      | Use local exhaust ventilation                          |
| Respiratory Protection                    | Use qualified respiratory protector if dust is present |
| Eye Protection                            | Wear safety glasses or goggles                         |
| Hand Protection                           | Wear chemically resistant gloves                       |
| Body Protection                           | Wear chemically resistant protective clothing          |

## 9. Physical and Chemical Properties

| Property    | Value         |
|-------------|---------------|
| Appearance  | Yellow paste  |
| Odor        | Not specified |
| pH          | No data       |
| Flash Point | No data       |
| Viscosity   | No data       |
| Solubility  | Not specified |

## 10. Stability and Reactivity

| Item                    | Information                                                              |
|-------------------------|--------------------------------------------------------------------------|
| Chemical Stability      | Unstable at room temperature                                             |
| Conditions to Avoid     | Heat, ignition sources, moisture, dust formation                         |
| Incompatible Materials  | Combustible materials, reducing agents                                   |
| Hazardous Decomposition | Irritative, corrosive, and toxic gases may be released during combustion |

## 11. Toxicological Information

| Effect                                        | Description                                                     |
|-----------------------------------------------|-----------------------------------------------------------------|
| Acute Toxicity                                | No data available                                               |
| Skin Irritation (Iron Oxides CI 77492)        | Irritation on skin                                              |
| Eye Irritation (Iron Oxides CI 77492)         | Irritation on eyes                                              |
| Respiratory Irritation (Iron Oxides CI 77492) | Irritation on respiratory system                                |
| Long-term Exposure                            | Respiratory system irritation may occur with prolonged exposure |

## 12. Ecological Information

| Category                      | Information       |
|-------------------------------|-------------------|
| Ecological Toxicity           | No data available |
| Persistency and Degradability | No data available |
| Bio-condensability            | No data available |

### 13. Disposal Considerations

| Item            | Information                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| Disposal Method | Dispose of contents/container in accordance with all applicable local, regional, national, and international regulations |

### 14. Transport Information

| Item                 | Information                |
|----------------------|----------------------------|
| Classification       | No Classification assigned |
| UN Number            | Not applicable             |
| Proper Shipping Name | Not applicable             |

### 15. Regulatory Information

| Item             | Information                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------|
| Regulatory Basis | Prepared in accordance with Article 41 of the Occupational Safety and Health Act and relevant KFDA regulations |

### 16. Other Information

| Item          | Information                                                                                                                                                                                                                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disclaimer    | To the best of our knowledge, the information contained herein is accurate. However, we cannot accept any responsibility or liability for any consequences arising from use of it. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. |
| Prepared by   | Darun Color Institute                                                                                                                                                                                                                                                                                                                                    |
| Document Type | For Educational Purposes Only                                                                                                                                                                                                                                                                                                                            |